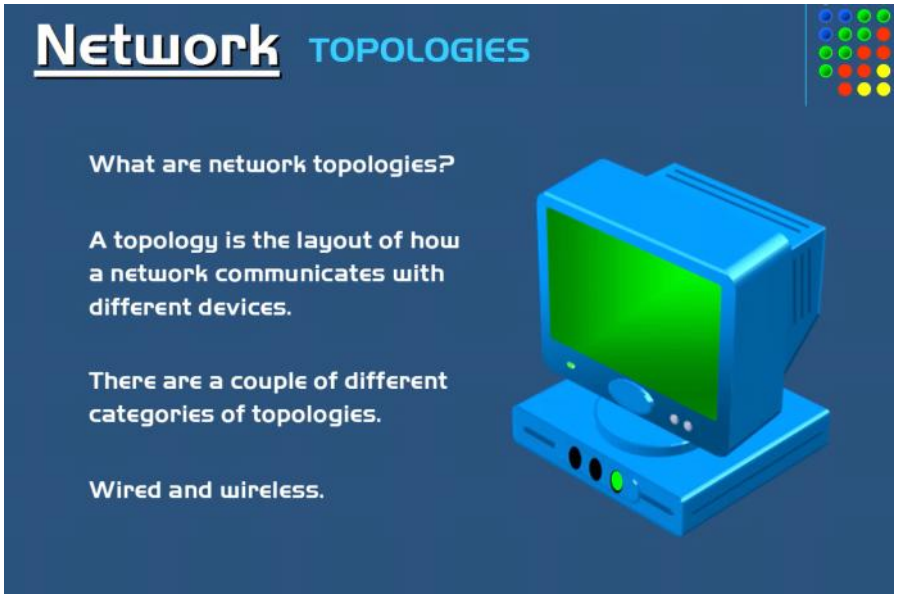
Network Topology

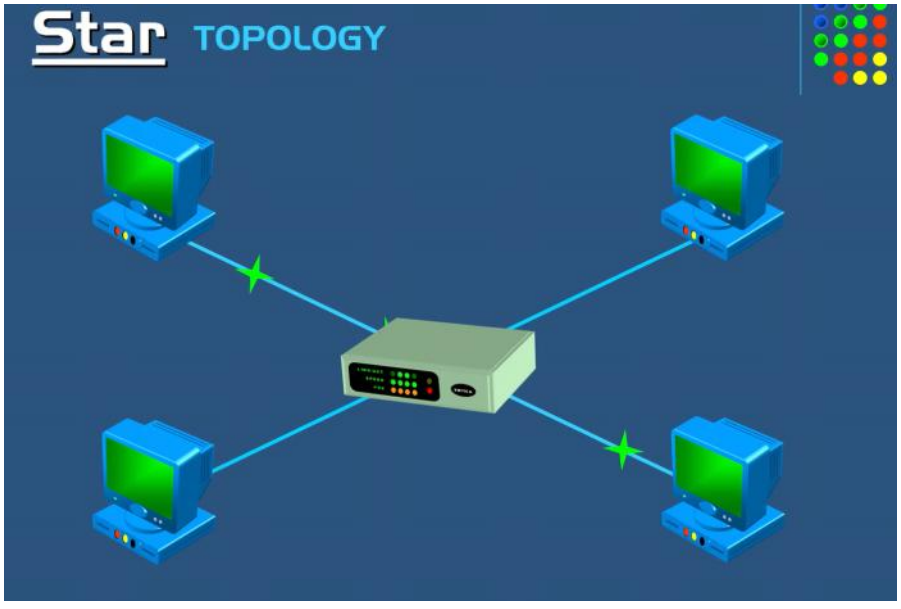
Sunday, March 15, 2026 5:06 PM

Network topology is the layout that shows how devices in a network are connected and communicate with each other. There are two main types: wired and wireless topologies.



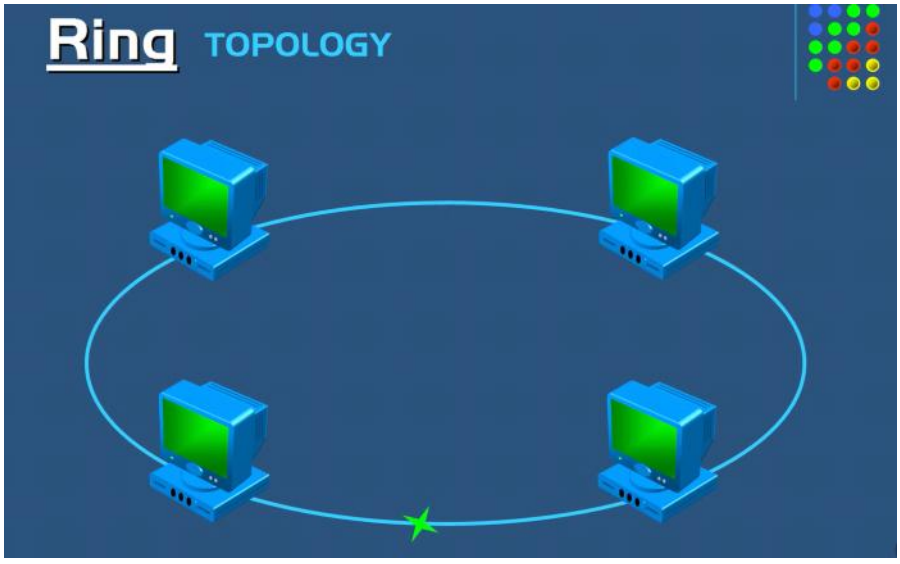
Star Topology

All devices connect to a central hub or switch. Data passes through this central device. If one cable or computer fails, others continue working. But if the central hub or switch fails, the entire network stops.



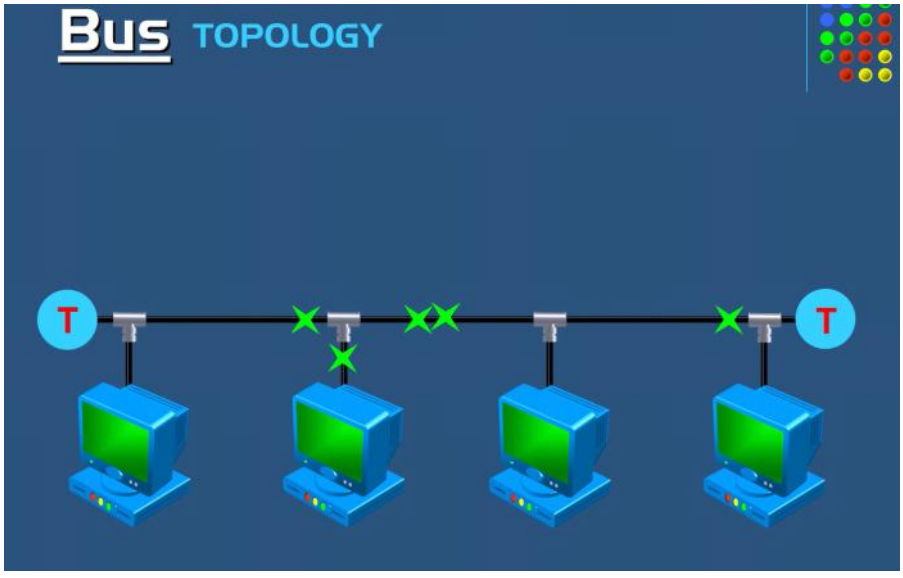
Ring Topology

Devices are connected in a closed loop. Each device has two neighbors and data travels around the ring until it reaches the destination. It is easy to install and troubleshoot, but if one device or cable fails the whole network is affected.



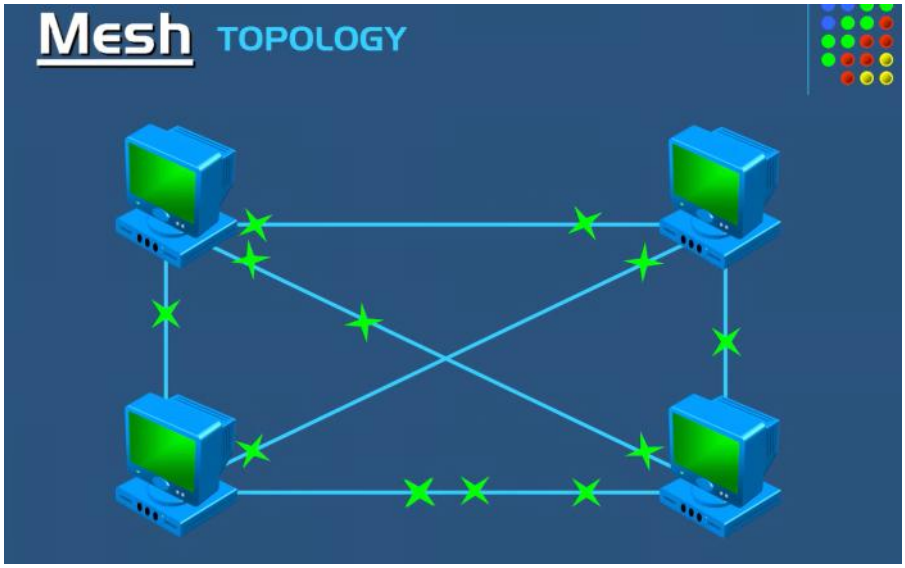
Bus Topology

All devices connect to a single backbone cable using BNC connectors. It is cheap and simple to implement. However the cable must be properly terminated at both ends, otherwise signal reflection can occur and disrupt communication.



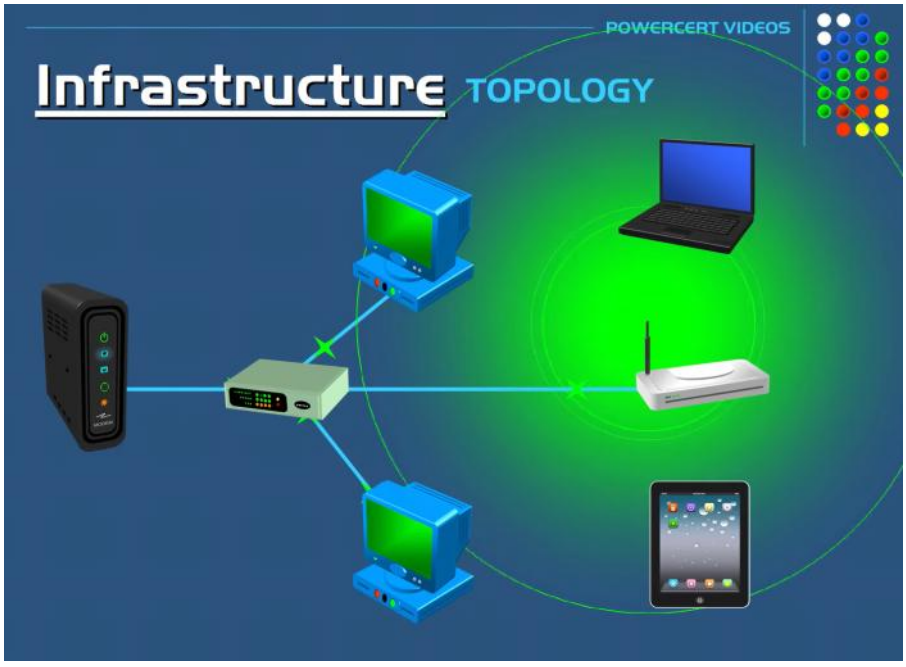
Mesh Topology

Each device connects to every other device. This provides high redundancy because data can travel through multiple paths if one connection fails. It requires a lot of cabling so it is expensive and mainly used in large networks like the internet.



Infrastructure Wireless Topology

Uses wired and wireless devices together. A wireless access point connects to a switch and allows wireless devices such as laptops and phones to join the network.



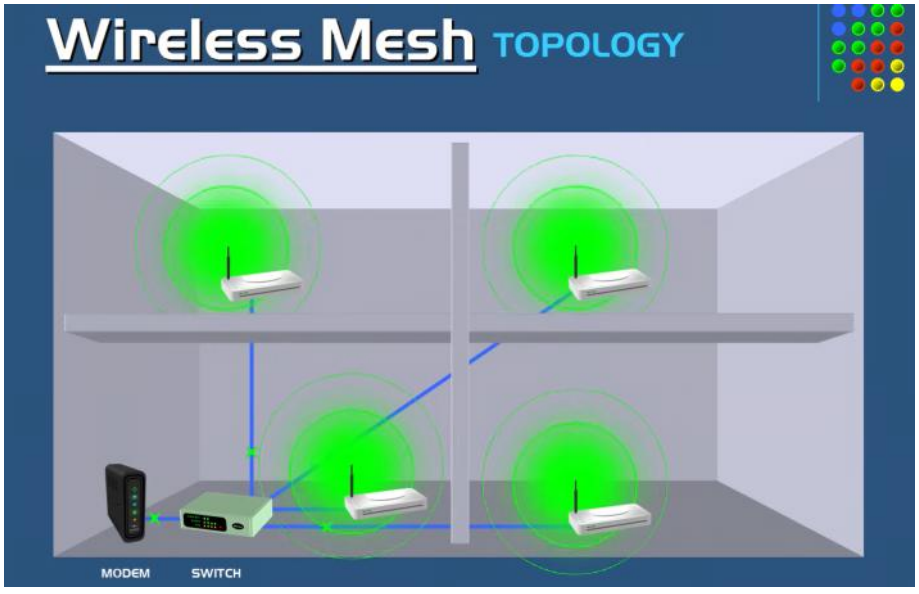
Ad Hoc Wireless Topology

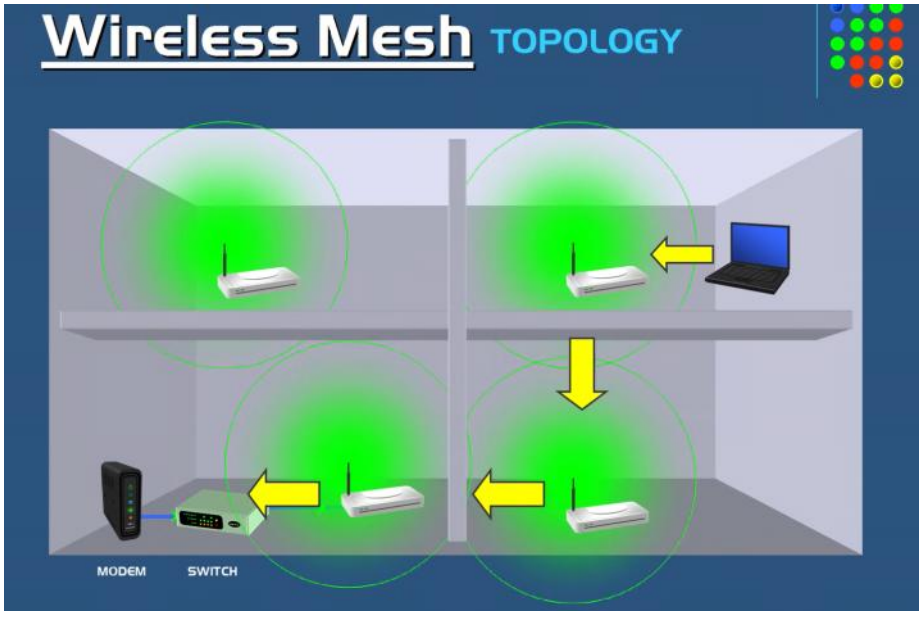
Devices connect directly to each other without routers, cables or access points. It forms a simple peer to peer wireless network and is useful for quick temporary connections.

Wireless Mesh Topology

Multiple wireless access points connect with each other wirelessly. Devices connect to the nearest access point and data is relayed between them. If one access point fails the network can reroute the data through another path, making it highly reliable.

Unlike the below one it doesn't require the cables shown in the last picture. Its very redundant as it will router through other path if any one of AP fails.




Wireless Mesh TOPOLOGY
MODEM
SWITCH